Kaili Cao

Eric and Wendy Schmidt AI in Science Postdoctoral Fellow, Michigan Institute for Data and AI in Society (MIDAS), University of Michigan, Ann Arbor, Michigan, United States

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Positions

2026 – 2028: **Schmidt AI in Science Postdoctoral Fellow**, University of Michigan, Ann Arbor, United States

2024 – present: **Member**, Asteroseismology Using The Galactic Bulge Time Domain Survey (Roman Space Telescope Wide-Field Science Team), Columbus, United States

2023 – present: **Member**, Maximizing Cosmological Science with the Roman High Latitude Imaging Survey (Roman Space Telescope Project Infrastructure Team), Pasadena, United States

2023 – 2026: **Graduate Research Assistant**, Center for Cosmology and AstroParticle Physics (CCAPP), The Ohio State University, Columbus, United States

2019: **Visiting Undergraduate Student**, Massachusetts Institute of Technology, Cambridge, United States

Education

2023 – 2026: **Doctor of Philosophy, Physics**, The Ohio State University, Columbus, United States
Dissertation: *Renovating Analysis Infrastructure for the Next Generation of Astronomical Surveys*

2021 – 2023: **Master of Science, Physics**, The Ohio State University, Columbus, United States

2020 – 2021: **31 credits, Computer Science and Technology**, Nanjing University, Nanjing, China

2016 – 2020: **Bachelor of Arts, French Language and Literature**, Nanjing University, Nanjing, China

Honors and Awards

2026 – 2028: **Eric and Wendy Schmidt AI in Science Postdoctoral Fellowship**, Michigan Institute for Data and AI in Society, University of Michigan.

2023 – 2026: Research support as a member of NASA's Roman HLIS Cosmology Project Infrastructure Team and the Roman asteroseismology Wide-Field Science Team (as a PhD student).

Research Interests

Cosmology: Weak Gravitational Lensing, Cosmic Shear Measurement, Parameter Inference, Fisher Forecasts

Stellar Physics: Stellar Models, Stellar Evolution Code, Interior Physics, Asteroseismology, Red Giant Branch

Methods: Numerical Simulations, Numerical Methods, Image Processing, Data Analysis, Machine Learning

Publications

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First-Author Publications

Oct 2026: **Kaili Cao**, David H. Weinberg, Vivian Miranda, et al., Fisher Forecasts for Cosmological Yields from 3×2pt Analysis of the Roman Space Telescope High Latitude Imaging Survey, *The Astrophysical Journal*, Volume 1010, Number 1, October 2026, Page 21, doi:10.3847/1538-4357/ae9bc0

Jul 2026: **Kaili Cao**, Efficient Optimal Image Reconstruction for the Nancy Grace Roman Space Telescope and Beyond: II. Implementation of Effortless, *arXiv e-prints*, arXiv:2607.09862 [astro-ph.IM], July 2026, doi:10.48550/arXiv.2607.09862 [GitHub]

Jul 2026: **Kaili Cao**, Efficient Optimal Image Reconstruction for the Nancy Grace Roman Space Telescope and Beyond: I. First Results with Effortless, *arXiv e-prints*, arXiv:2607.06674 [astro-ph.IM], July 2026, doi:10.48550/arXiv.2607.06674 [GitHub]

Mar 2026: **Kaili Cao** and Roman HLIS Cosmology PIT, Linear Image Regridding and Coaddition with Oversampled Point Spread Functions: Lessons from 1D, *The Astronomical Journal*, Volume 171, Number 3, March 2026, Page 140, doi:10.3847/1538-3881/ae356a [GitHub]

Feb 2026: **Kaili Cao**, Christopher M. Hirata, Katherine Laliotis, et al., Simulating Image Coaddition with the Nancy Grace Roman Space Telescope. IV. Hyperparameter Optimization and Experimental Features, *The Astrophysical Journal*, Volume 998, Number 2, February 2026, Page 304, doi:10.3847/1538-4357/ae3a97 [GitHub]

Jan 2026: **Kaili Cao** and Marc H. Pinsonneault, Modeling APOKASC-3 Red Giants. I. The First Dredge-up and Red Giant Branch Bump, *The Astrophysical Journal*, Volume 997, Number 1, January 2026, Page 120, doi:10.3847/1538-4357/ae22d1 [GitHub, Zenodo]

Dec 2025: **Kaili Cao**, ContEvol Formalism: Numerical Methods Based on Hermite Spline Optimization, *Mathematics*, Volume 13, Issue 24, December 2025, Article Number 3981, doi:10.3390/math13243981 [GitHub]

Sep 2025: **Kaili Cao** and Marc H. Pinsonneault, Toward Accurate Asteroseismic Masses for Luminous Giants, *The Astrophysical Journal Letters*, Volume 990, Number 1, September 2025, Page L26, doi:10.3847/2041-8213/adfa12

Apr 2025: **Kaili Cao**, Christopher M. Hirata, Katherine Laliotis, et al., Simulating Image Coaddition with the Nancy Grace Roman Space Telescope. III. Software Improvements and New Linear Algebra Strategies, *The Astrophysical Journal Supplement Series*, Volume 277, Number 2, April 2025, Page 55, doi:10.3847/1538-4365/adb580 [GitHub]

May 2021: **Kaili Cao**, David J. Barnes, and Mark Vogelsberger, Studying galaxy cluster morphological metrics with MOCK-X, *Monthly Notices of the Royal Astronomical Society*, Volume 503, Issue 3, May 2021, Pages 3394–3413, doi:10.1093/mnras/stab605

Coauthored Publications in Cosmology

Jul 2026: Chun-Hao To, Tae-Hyeon Shin, Chihway Chang, et al. (with **Kaili Cao** as 10th of 21 authors), Robust Photometry for Roman High-Latitude Imaging Survey Cosmology Using Roman and Rubin Imaging, *arXiv e-prints*, arXiv:2607.09849 [astro-ph.IM], July 2026, doi:10.48550/arXiv.2607.09849

Jun 2026: Junzhou Zhang, Chihway Chang, Jiachuan Xu, et al. (with **Kaili Cao** as 7th of 9 authors), Accurate modeling for 3×2pt analyses in Roman and Rubin: a study of model approximations, *arXiv e-prints*, arXiv:2606.23781 [astro-ph.CO], June 2026, doi:10.48550/arXiv.2606.23781

Apr 2026: Katherine Laliotis, Christopher M. Hirata, Emily Macbeth, and **Kaili Cao**, Removing correlated noise stripes from the Nancy Grace Roman Space Telescope survey images, *Publications of the Astronomical Society of Japan*, Volume 78, Issue 2, April 2026, Page 810–823, doi:10.1093/pasj/psag020 [GitHub]

Dec 2025: OpenUniverse, The LSST Dark Energy Science Collaboration, The Roman HLIS Project Infrastructure Team, et al. (authors in alphabetical order, including **K. Cao**), OpenUniverse2024: a shared, simulated view of the sky for the next generation of cosmological surveys, *Monthly Notices of the Royal Astronomical Society*, Volume 544, Issue 4, December 2025, Pages 3799–3823, doi:10.1093/mnras/staf1833

Dec 2024: Katherine Laliotis, Emily Macbeth, Christopher M. Hirata, et al. (with **Kaili Cao** as 4th of 6 authors), Analysis of Biasing from Noise from the Nancy Grace Roman Space Telescope: Implications for Weak Lensing, *Publications of the Astronomical Society of the Pacific*, Volume 136, Number 12, December 2024, Page 124506, doi:10.1088/1538-3873/ad9bec

Mar 2024: Masaya Yamamoto, Katherine Laliotis, Emily Macbeth, et al. (with **Kaili Cao** as 7th of 19 authors), Simulating image coaddition with the Nancy Grace Roman Space Telescope - II. Analysis of the simulated images and implications for weak lensing, *Monthly Notices of the Royal Astronomical Society*, Volume 528, Issue 4, March 2024, Pages 6680–6705, doi:10.1093/mnras/stae177

Feb 2024: Christopher M. Hirata, Masaya Yamamoto, Katherine Laliotis, et al. (with **Kaili Cao** as 7th of 19 authors), Simulating image coaddition with the Nancy Grace Roman Space Telescope - I. Simulation methodology and general results, *Monthly Notices of the Royal Astronomical Society*, Volume 528, Issue 2, February 2024, Pages 2533–2561, doi:10.1093/mnras/stae182

Sep 2021: David J. Barnes, Mark Vogelsberger, Francesca A. Pearce, et al. (with **Kaili Cao** as 6th of 8 authors), Characterizing hydrostatic mass bias with MOCK-X,

Monthly Notices of the Royal Astronomical Society, Volume 506, Issue 2, September 2021, Pages 2533–2550, doi:10.1093/mnras/stab1276

Coauthored Publications in Stellar Astrophysics

Aug 2026: Marc H. Pinsonneault, Jennifer L. van Saders, Lyra Cao, et al. (with **Kaili Cao** as 13th of 14 authors), The YREC Stellar Evolution Code: Public Data Release, *The Astrophysical Journal*, Volume 1006, Number 2, August 2026, Page 220, doi:10.3847/1538-4357/ae71cf [GitHub]

Oct 2025: B. Liagre, R. A. García, S. Mathur, et al. (with **K. Cao** as 7th of 12 authors), Beyond the Nyquist frequency: Asteroseismic catalog of undersampled Kepler late subgiants and early red giants, *Astronomy & Astrophysics*, Volume 702, October 2025, Article Number A144, doi:10.1051/0004-6361/202555167

Jul 2025: Trevor J. Weiss, Noah J. Downing, Marc H. Pinsonneault, et al. (with **Kaili Cao** as 7th of 13 authors), Modeling Asteroseismic Yields for the Roman Galactic Bulge Time-domain Survey, *The Astrophysical Journal*, Volume 987, Number 2, July 2025, Page 181, doi:10.3847/1538-4357/adde5b

May 2025: M. Vrard, M. H. Pinsonneault, Y. Elsworth, et al. (with **K. Cao** as 11th of 15 authors), Red giant evolutionary status determination: The complete Kepler catalog, *Astronomy & Astrophysics*, Volume 697, May 2025, Article Number A165, doi:10.1051/0004-6361/202452635

Feb 2025: Marc H. Pinsonneault, Joel C. Zinn, Jamie Tayar, et al. (with **Kaili Cao** as 19th of 41 authors), APOKASC-3: The Third Joint Spectroscopic and Asteroseismic Catalog for Evolved Stars in the Kepler Fields, *The Astrophysical Journal Supplement Series*, Volume 276, Number 2, February 2025, Page 69, doi:10.3847/1538-4365/ad9fef

May 2024: John D. Roberts, Marc H. Pinsonneault, Jennifer A. Johnson, et al. (with **Kaili Cao** as 11th of 19 authors), Nature versus nurture: distinguishing effects from stellar processing and chemical evolution on carbon and nitrogen in red giant stars, *Monthly Notices of the Royal Astronomical Society*, Volume 530, Issue 1, May 2024, Pages 149–166, doi:10.1093/mnras/stae820

Selected Presentations

2026

Jul 2026: **Fisher Forecasts for Cosmological Yields from 3×2pt Analysis of the Roman Space Telescope High Latitude Imaging Survey (poster)**, *Shifting Landscapes in Astrophysics: New Frontiers to Explore with Roman*, Pasadena, United States. [Poster, Lightning talk video, Lightning talk slides]

Jul 2026: **Effortless: Efficient Optimal Image Reconstruction for Roman and Beyond (poster)**, *Shifting Landscapes in Astrophysics: New Frontiers to Explore with Roman*, Pasadena, United States. [Poster, Lightning talk video, Lightning talk slides]

Jun 2026: **Renovating Analysis Infrastructure for the Next Generation of Astronomical Surveys (dissertation defense)**, *The Ohio State University*, Columbus, United States. [Slides]

Apr 2026: **Weak Gravitational Lensing Cosmology with the Roman Space Telescope High Latitude Imaging Survey (seminar)**, *Cosmology Group Seminar (University of Michigan)*, Ann Arbor, United States. [Slides]

Mar 2026: **Hot-wiring the Roman Sky: Resolving the Undersampling Crisis with Fast IMCOM for Real-Time Discovery (contributed talk)**, *RAPID Response: Hot-wiring the Next Generation of Time-Domain Science (Caltech/IPAC)*, Pasadena, United States. (Fast IMCOM is the earlier name of Effortless.) [Slides]

Jan 2026: **Fisher Forecasts for Cosmological Yields from 3×2pt Analysis of the Roman Space Telescope High Latitude Imaging Survey (poster)**, *Arizona Winter School on Multi-Probe Cosmology in the Roman ST Era (University of Arizona)*, Tucson, United States. [Poster, an earlier version of the July 2026 poster]

Jan 2026: **Fast IMCOM: Unlocking New Scientific Frontiers with High-Fidelity Image Coaddition (contributed oral presentation)**, *247th Meeting of the American Astronomical Society*, Phoenix, United States. (Fast IMCOM is the earlier name of Effortless.) [Slides, Abstract]

2025

Nov 2025: **Early Career Scientist Talk: Stage IV Observation, Stage IV Computation**, *Roman HLIS Cosmology PIT Face-to-Face Meeting 2025 (University of Pennsylvania)*, Philadelphia, United States. [Slides]

Oct 2025: **Testing Stellar Theory with Joint Spectroscopic and Asteroseismic Catalogs (invited seminar talk)**, *AstroCoffee (University of Pittsburgh)*, Pittsburgh, United States. [Slides]

Oct 2025: **Forecasts for weak lensing cosmology with Roman Space Telescope (contributed parallel talk)**, *28th International Conference on Particle Physics and Cosmology (COSMO-25)*, Pittsburgh, United States. [Slides]

Oct 2025: **Toward Accurate Asteroseismic Masses for Luminous Giants (invited talk, online)**, *Galactic Genesis Working Group (SDSS-V Milky Way Mapper)*. [Slides]

Aug 2025: **PyIMCOM Hack Session 2025**, *Shear and Clustering Measurement Working Group Hack Week (Roman HLIS Cosmology PIT)*, Columbus, United States. [Slides]

2024 and Before

Jul 2024: **IMCOM: optimized image coaddition pipeline for Roman weak lensing cosmology (contributed talk)**, *Challenging Theory with Roman: From Planet Formation to Cosmology*, Pasadena, United States. [Abstract (please search for "IMCOM"), Slides, Video]

Sep 2022: **Theoretical predictions concerning the first dredge-up (presentation)**, *Summer Research Poster Session (The Ohio State University)*, Columbus, United States. [Slides]

Aug 2019: **How to define a relaxed cluster?**, *Summer MKI Undergraduate Research Forum (SMURF; Massachusetts Institute of Technology)*, Cambridge, United States. [Slides]

Teaching Experience

2025: **Microteaching Lesson, College Teaching (ESEPSY 7404)**, The Ohio State University, Columbus, United States
Planned a lesson by backward design, "Basic Counting: Permutations and Combinations," and taught it to my peers. [Slides]

2022 – 2023: **Instructor, Online Python Tutorial**
About 46 students in 4 cohorts. Gave lectures, designed exercises, and mentored personal projects.

2021 – 2023: **Teaching Associate (Grader), Physics 7501 & 7502: (Graduate) Quantum Mechanics**, The Ohio State University, Columbus, United States

Miscellany: Taught or tutored college mathematics (2017, 2018, 2020, 2023), college physics (2023), graduate school application (2022, 2023, 2024), French phonology (2017, 2018), etc.

Service

2026 – present: **Lead**, MapHub (the MIDAS arXiv Preprint Hub), a carpentry of the Schmidt AI in Science Fellowship, with Kyle Heyblom, Eva Albalghiti and Dimitrios Simatos.

2025: **Organizer**, PyIMCOM hack session for the Roman HLIS Cosmology Project Infrastructure Team.

Skills

Programming: Python (main), C/C++, Fortran, Shell/Perl, Wolfram Mathematica

Languages: Chinese (native), English (working proficiency), French (undergraduate major)

Certificates: Productivity Program (The Ohio State University, 2021), Responsible Conduct of Research (The Ohio State University, 2024), Data Science Boot Camp (The Erdős Institute, 2025)

References

Available on request.